

Finite Element Groundwater Seepage

Phase2 has the capability to carry out a finite element groundwater seepage analysis, for saturated / unsaturated, steady state flow conditions.

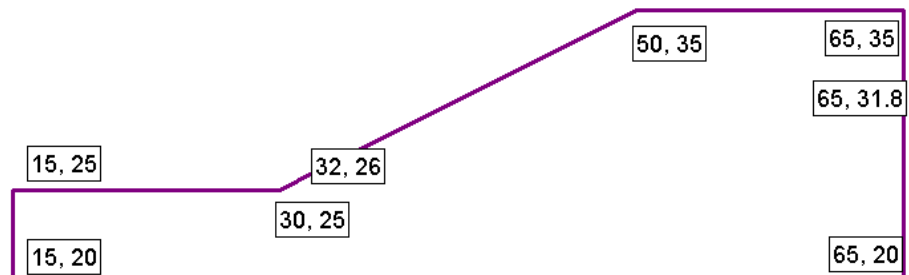
This tutorial will demonstrate the basic features of performing a groundwater seepage analysis with *Phase2*, and how this functionality is fully integrated with the stress analysis functionality of *Phase2*.

- After a groundwater seepage analysis is computed, the results (pore pressures), are automatically utilized in the *Phase2* stress analysis to calculate effective stress.
- The seepage analysis capability in *Phase2* can also be used as a standalone groundwater program, independently of the stress analysis functionality of *Phase2*.

Topics Covered

- Hydraulic boundary conditions
- Hydraulic material properties
- Ponded water loading
- Groundwater + Stress Analysis
- Computed Water Table
- Flow Vectors
- Flow Lines
- Iso-Lines
- Discharge Sections

Geometry



Model

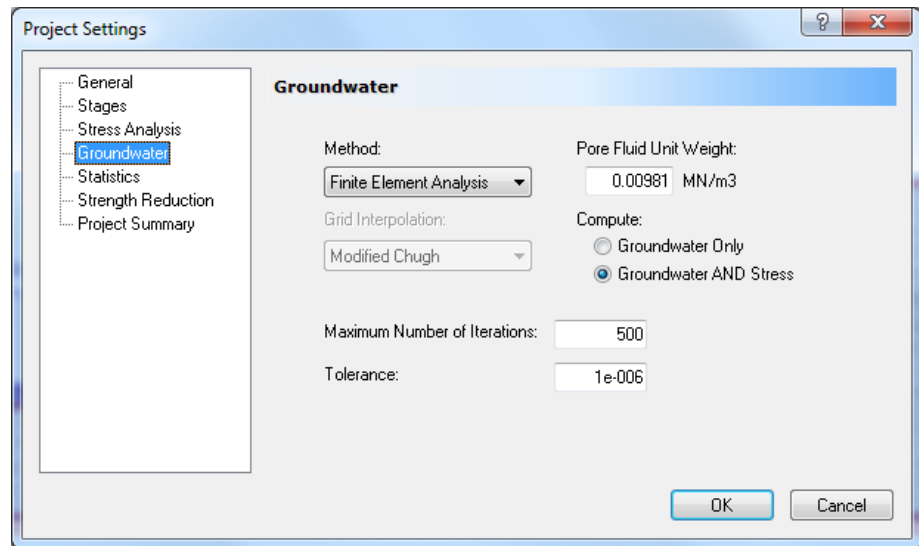
Start the *Phase2* Model program.

Project Settings

The first thing you must do before you can start defining a finite element groundwater model, is to set the Groundwater Method = Finite Element Analysis in Project Settings.



Select Project Settings from the Analysis menu, select the Groundwater tab, and set the *Method* = *Finite Element Analysis*.



Also, select the General tab and make sure that the Units are set to *Metric, stress as MPa*, as that is what we will be using for this tutorial. Select OK.

External Boundary



This model only requires an External boundary to define the geometry. Select the Add External option from the Boundary menu, and enter the following coordinates in the prompt line at the bottom right of the screen.

```
Enter vertex [t=table,i=circle,esc=cancel]: 15 20
Enter vertex [...]: 65 20
Enter vertex [...]: 65 31.8
Enter vertex [...]: 65 35
Enter vertex [...]: 50 35
Enter vertex [...]: 32 26
Enter vertex [...]: 30 25
Enter vertex [...]: 15 25
Enter vertex [...,c=close,esc=cancel]: c
```

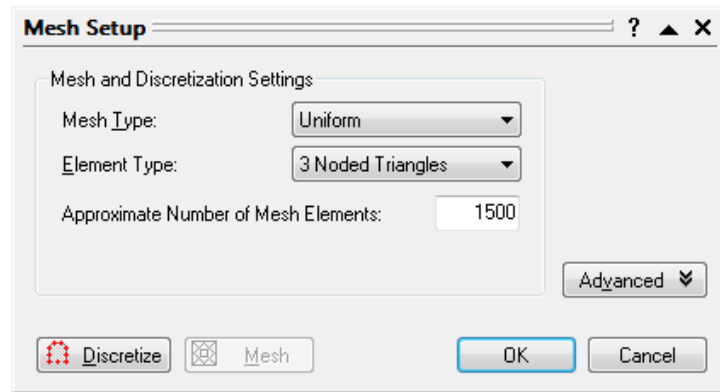
Entering **c** at the last prompt automatically closes the boundary and exits the Add External Boundary option.

Select Zoom All from the toolbar (or press the F2 function key) to zoom the model to the center of the view.

Mesh



Now generate the finite element mesh. Select the **Mesh Setup** option in the **Mesh** menu. Change the Mesh Type to Uniform. Leave the default element type (3 Noded Triangles) and the number of elements (1500). Click the Discretize button followed by the Mesh button.



Close the Mesh Setup dialog by selecting the OK button.

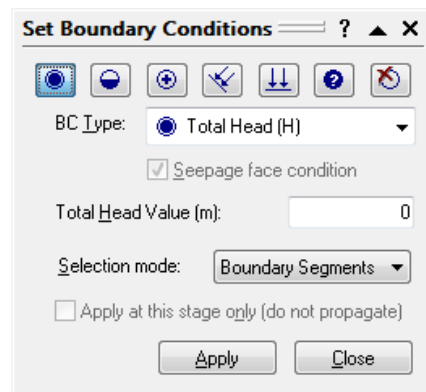
Hydraulic Boundary Conditions



Now define the hydraulic boundary conditions. Select the Set Boundary Conditions option from the toolbar or the Groundwater menu.

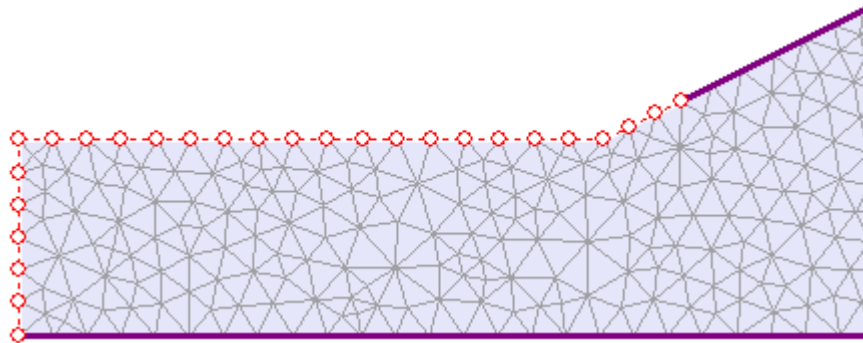
NOTE: the stress analysis boundary conditions are automatically hidden when you are defining groundwater boundary conditions.

You will see the Set Boundary Conditions dialog, which allows you to define the hydraulic boundary conditions for the groundwater analysis.



Let's first set the Total Head boundary conditions:

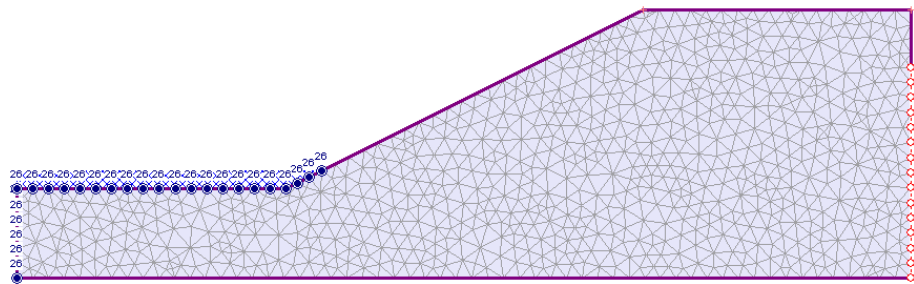
1. Make sure the Total Head boundary condition option is selected in the Set Boundary Conditions dialog, as shown above.
2. In the dialog, enter a Total Head Value = 26 meters. Also make sure the Selection Mode is set to Boundary Segments.
3. Now you must select the desired boundary segments, by clicking on them with the mouse.
4. Click on the **THREE** segments of the external boundary indicated in the following figure. (i.e. the left edge of the external boundary, and the two segments at the toe of the slope).



5. When the segments are selected, right-click the mouse and select Done Selection. A boundary condition of Total Head = 26 meters is now assigned to these line segments.

NOTE: the hatch pattern represents ponded water which is defined by the Total Head boundary condition of 26 meters on the selected segments.

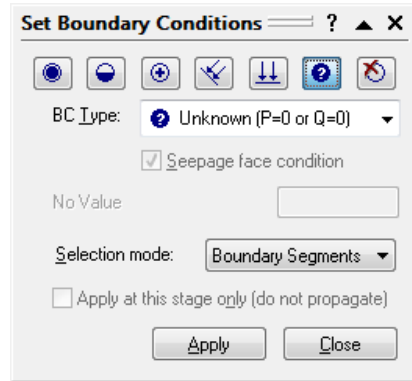
6. Now enter a Total Head Value = 31.8 meters in the dialog. Select the lower right segment of the external boundary, as shown below. Right-click and select Done Selection.



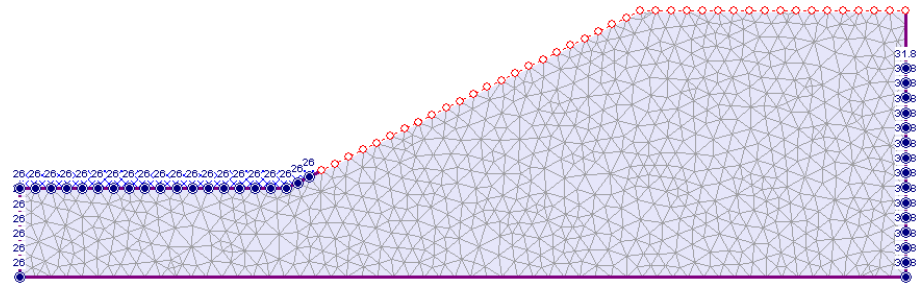
The Total Head boundary conditions represent the elevation of the phreatic surface (ponded water) at the left of the model (26 m), and the elevation of the phreatic surface at the right edge of the model (31.8 m).

Now we need to assign the Unknown ($P=0$ or $Q=0$) boundary condition to the upper two segments of the slope.

7. In the Set Boundary Conditions dialog, select the Unknown ($P=0$ or $Q=0$) boundary condition option.



8. Select the upper two segments of the slope, as shown below. Right-click and select Done Selection.



9. The necessary hydraulic boundary conditions are now assigned.

Stress Analysis Boundary Conditions

Although this tutorial is primarily concerned with how to define a groundwater seepage analysis model, we will also discuss the stress analysis aspects of the model, since in many cases you will be performing both a groundwater and a stress analysis.

When you selected the Set Boundary Conditions option (in the previous section), the stress analysis boundary conditions were automatically hidden.



To restore the display of the stress analysis boundary conditions, select the Show Boundary Conditions option from the toolbar or the Groundwater menu.

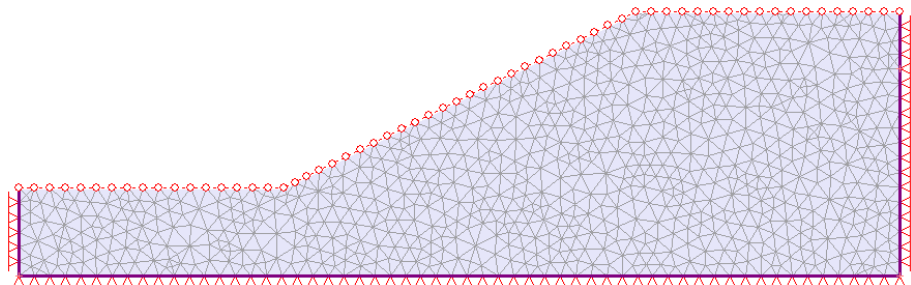
NOTE: the Show Boundary Conditions option can be selected at any time to toggle the display between stress analysis and groundwater boundary conditions.

Notice that the stress analysis boundary conditions (in this case, Fixed X,Y conditions on the external boundary) are now displayed.

First we have to free the segments of the external boundary representing the slope surface.



1. Select the Free option from the toolbar or the Displacements menu.
2. Select the four line segments defining the ground surface of the slope, as shown below.



3. Right-click and select Done Selection.

The slope surface is now free, however, this process has also freed the vertices at the upper left and upper right corners of the model. Since these edges should be restrained, we have to make sure that these two corners are restrained.

We can use a right-click shortcut to assign boundary conditions:

4. Right-click the mouse directly on the vertex at (15,25). From the popup menu select the Restrain X,Y option.
5. Right-click the mouse directly on the vertex at (65,35). From the popup menu select the Restrain X,Y option.

The restraint boundary conditions are now correctly applied.

Ponded Water Load

An important point to remember when you are defining a *Phase2* model which includes Ponded Water, and you are carrying out both a groundwater seepage analysis and a stress analysis:

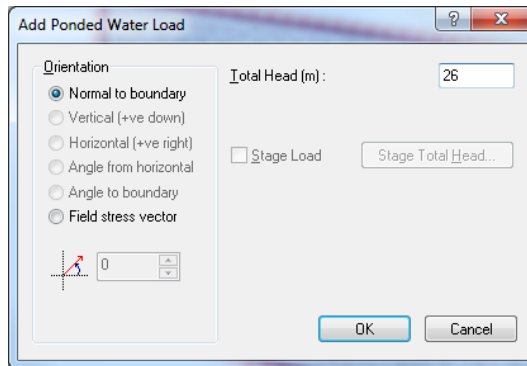
- The **weight** of the Ponded Water must be defined by adding a Ponded Water distributed load to the model.

The total head boundary conditions that you use to define the hydraulic boundary conditions DO NOT define the weight of the ponded water. Conversely, the Ponded Water distributed load DOES NOT define the total head boundary conditions required by the groundwater analysis.

The Ponded Water load is defined as follows:



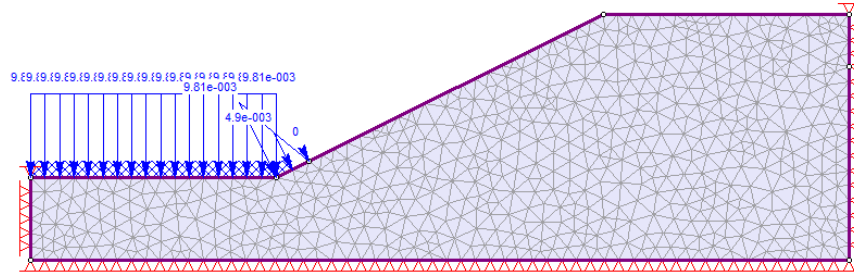
1. Select the Add Ponded Water Load option from the toolbar or the Distributed Loads sub-menu of the Loading menu.
2. You will see the Add Ponded Water Load dialog. Enter a Total Head value of 26 meters and select OK.



3. Select the slope segment between the vertices at (15,25) and (30,25) and the slope segment between vertices (30,25) and (32,26). Right-click and select Done Selection.
4. The ponded water load will be added to the model, and is represented by blue arrows applied normal to the selected boundary segments.

NOTE: *Phase2* automatically determines the magnitude of the load based on the value of Total Head, the elevation of the line segments, and the unit weight of water entered in the Project Settings dialog.

5. Your model should appear as in the following figure.



Field Stress



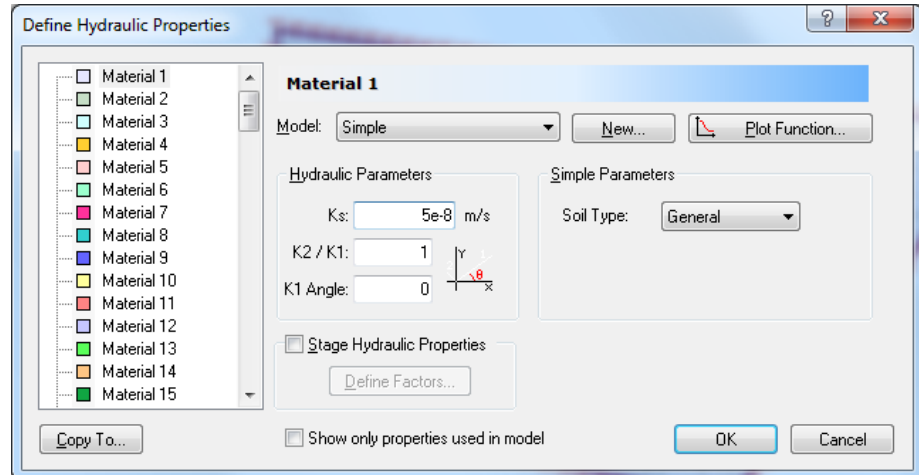
A surface model usually requires Gravity field stress, so let's quickly define that. Select the Field Stress option from the Loading menu, select the Gravity field stress option, and also select the Use Actual Ground Surface checkbox. Select OK.

Define Hydraulic Properties

Now define the hydraulic properties (permeability) of the slope material.



Select the **Define Hydraulic** option from the toolbar or the Properties menu.



In the Define Hydraulic Properties dialog, enter a saturated permeability $K_s = 5e-8$. Select OK.

NOTE: since we are dealing with a single material model, and since you entered properties with the first (default) tab selected, you do not have to Assign these properties to the model. The properties are automatically assigned by *Phase2*.

Define Material Properties

Since we are primarily concerned with the groundwater analysis for this tutorial, we will just use the default material strength and stiffness properties in the Define Material Properties dialog.

Discharge Section

A Discharge Section allows you to compute the steady-state, volumetric flow rate through a user-defined line segment. Let's add a Discharge Section to the model.



1. Select Add Discharge Section from the toolbar or the Groundwater menu.
2. Right-click the mouse and make sure that the Snap options are enabled (checkbox is displayed beside each option).
3. Click the mouse on the vertex at the crest of the slope at (50,35).
4. Click the mouse at the point (50,20) on the lower edge of the external boundary to create a vertical discharge section between the crest of the slope and the lower edge of the model.

Compute

Now save the model. Select Save from the toolbar and use the Save As dialog to save the file. You are now ready to run the analysis.

To compute the groundwater seepage analysis, you have two choices:

Compute Groundwater Only



If you only wish to compute the groundwater analysis, without computing the stress analysis, then you can select the Compute (Groundwater Only) toolbar button. This is useful if you wish to check groundwater results before running the stress analysis, or if you are not interested in the stress analysis.

Compute Groundwater and Stress



If you select the main Compute option, then the groundwater seepage analysis will be computed first, and the stress analysis will be computed next. The stress analysis will utilize the pore pressures calculated from the groundwater analysis.

Select the main Compute option, so that both groundwater and stress analysis results will be calculated. The analysis should only take a few seconds.

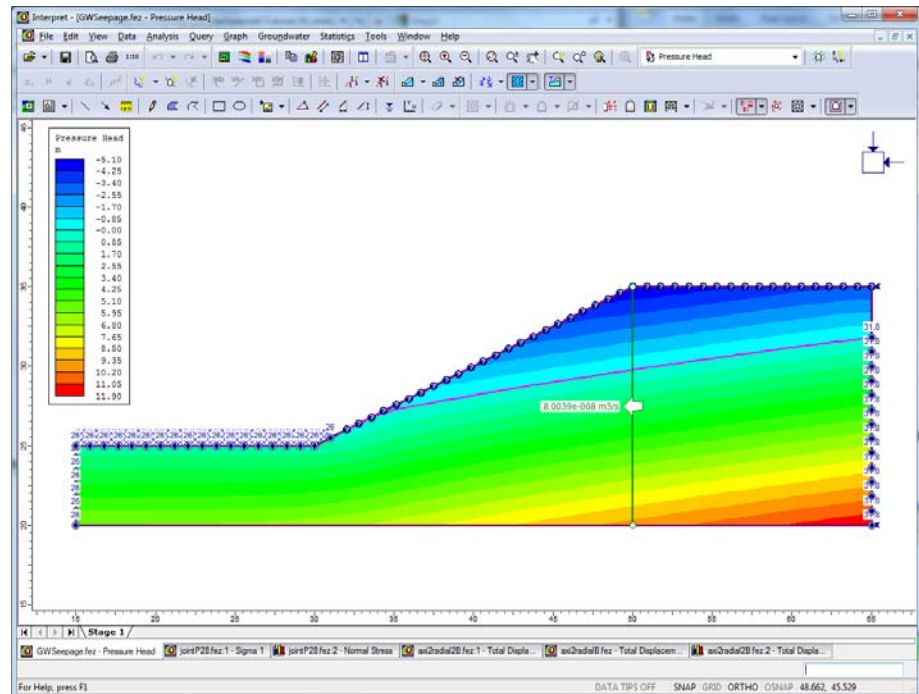
Interpret

We can now view the results of the groundwater and stress analysis in the *Phase2* Interpret program.



Select: Analysis → Interpret

Your screen should appear as follows.



Change the view so that you are viewing Pressure Head contours. The Legend in the upper left corner of the view, indicates the values of the contours.

The contour display can be customized with the Contour Options dialog, which is available in the toolbar, the View menu, or the right-click menu.

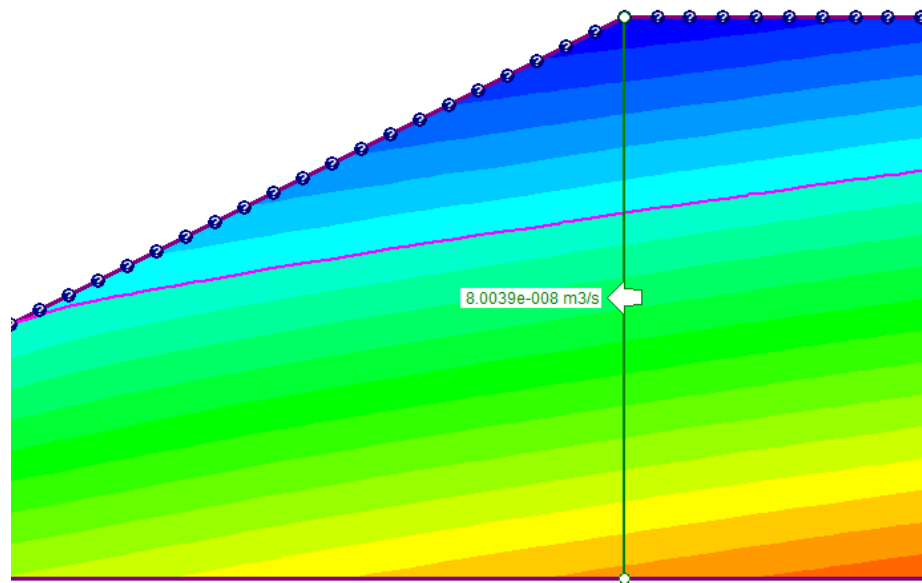
Also note that by default the groundwater boundary conditions are displayed (Total Head etc).

TIP: you can turn off the display of the Total Head values in the Display Options dialog (select the Groundwater tab and turn off the Show BC Values checkbox). This is also available as a toolbar shortcut.

Discharge Section

The Discharge Section (the vertical green line segment) displays the steady-state, volumetric flow rate of water, normal to the plane of the discharge section.

The flow rate is approximately $8\text{e-}8 \text{ m}^3/\text{s}$ across the discharge section, in the direction indicated by the arrow.



Display of flow across Discharge Section

The display of Discharge Sections can be turned on or off in the Display Options dialog or the toolbar. You can also use a right-click shortcut. Right-click on the Discharge Section and select Hide All Discharge Sections from the popup menu, to hide the discharge section.

Water Table

You will notice on the plot, a pink line which is displayed on the model. This line highlights the location of the Pressure Head = 0 contour boundary.



By definition, a Water Table is defined by the location of the Pressure Head = 0 contour boundary. Therefore, for a slope model such as this, this line represents the position of the Water Table (phreatic surface) determined from the finite element analysis.

The display of the Water Table can be turned on or off using the toolbar shortcut, the Display Options dialog, or the right-click shortcut (right-click ON the Water Table and select Hide Water Table).

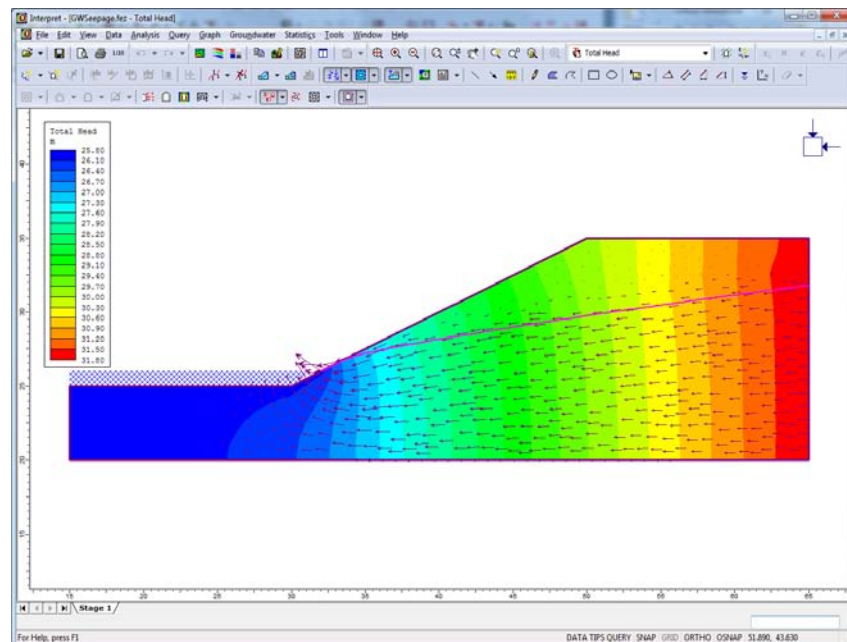
Notice that the contours of Pressure Head, above the Water Table, have negative values. The negative pressure head calculated above the water table, is commonly referred to as the “matric suction” in the unsaturated zone. This is discussed later in the tutorial.

To view contours of other data (Total Head, Pressure, or Discharge Velocity), simply use the mouse to select from the drop-down list in the toolbar.

Select Total Head contours from the drop-down list.

Flow Vectors

Right click the mouse and select Display Options. Select the Groundwater tab. Toggle ON the Flow Vectors option. Toggle OFF all of the Boundary Condition options. Select Done. (Flow Vectors and other Display Options can also be toggled on or off with shortcut buttons in the toolbar.)



Total Head contours and flow vectors.

As expected, the direction of the flow vectors corresponds to decreasing values of the total head contours.

NOTE: the relative size of the flow vectors (as displayed on the screen), corresponds to the magnitude of the flow velocity. Select Total Discharge Velocity contours (from the toolbar list), and verify this. The size of the flow vectors can be scaled in the Display Options dialog. This is left as an optional exercise.



Turn off the flow vectors by re-selecting the flow vectors option from the toolbar.

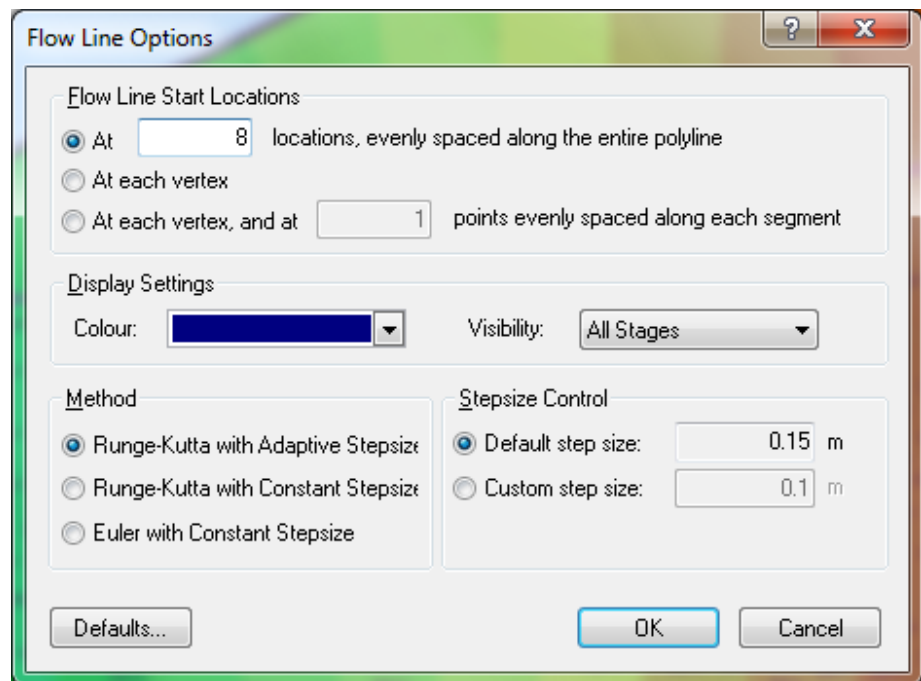
Flow Lines

Select Total Head contours again.

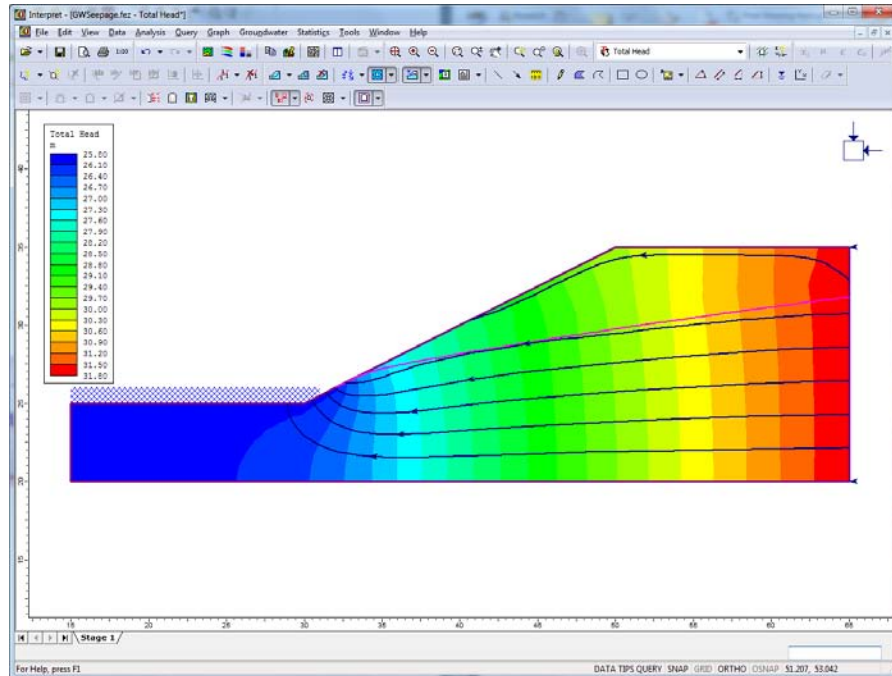
We can also add Flow Lines to the plot. Flow lines can be added individually, with the Add Flow Line option. Or multiple flow lines can be automatically generated with the Add Multiple Flow Lines option. Let's do that.



1. Select **Add Multiple Flow Lines** from the toolbar or the Groundwater menu.
2. Make sure the Snap option is enabled in the Status Bar. If not, then right click the mouse and enable Snap from the popup menu, or click on the word Snap in the Status Bar.
3. Click the mouse on the upper right corner of the external boundary, i.e., the vertex at (65,35).
4. Click the mouse on the lower right corner of the external boundary, i.e., the vertex at (65,20).
5. Right click and select Done.
6. You will then see a dialog. Enter a value of 8 and select OK.



The generation of the flow lines may take a few seconds. Your screen should look as follows.



Total Head contours and flow lines

Notice that the flow lines are perpendicular to the Total Head contours. (Note: only 6 flow lines are displayed, although we entered a value of 8, because the first and last flow lines are exactly on the boundary, and are not displayed.)



Now delete the flow lines. Select Delete Flow Lines from the toolbar, right click and select Delete All, and select OK in the dialog which appears.

TIP: Flow Lines (and Iso-Lines, discussed in the next section) can be saved by selecting the Save Tools and Lines option. This will save all drawing tools, Flow Lines and Iso-lines, so that you don't have to re-create them each time you open a file in Interpret.

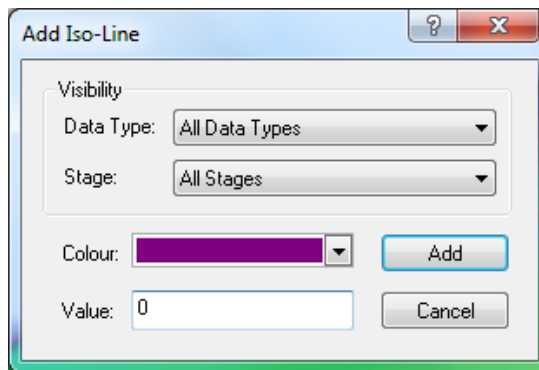
Iso-Lines

An iso-line is a line of constant contour value, displayed on a contour plot.

As we discussed earlier, the pink line which is displayed on the model, represents the Water Table determined by the groundwater analysis. By definition, the Water Table represents an iso-line of zero pressure head. Let's verify that the displayed Water Table does in fact represent a line of zero pressure ($P = 0$ iso-line), by adding an iso-line to the plot.



1. First, make sure you select Pressure Head contours.
2. Select the **Add Iso-Line** option from the toolbar, or the Iso-Line sub-menu in the Analysis menu.
3. Click the mouse on the Water Table line. You will then see the Add Iso-Line dialog.



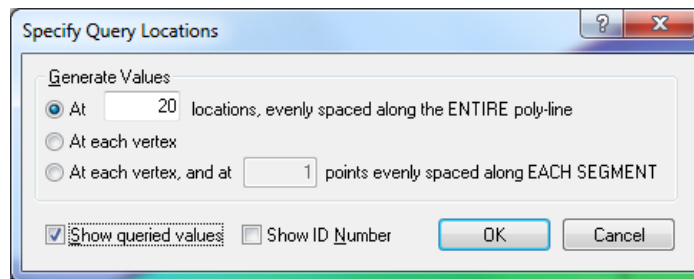
4. The dialog will display the exact value (Pressure Head) of the location at which you clicked. It may not be exactly zero, so enter zero in the dialog, and select the Add button.
5. An Iso-Line of zero pressure head, will be added to the model. It overlaps the displayed Water Table exactly, verifying that the Water Table is the $P = 0$ line.
6. Press Escape or right-click and select Cancel to exit the Add Iso-Line option.

Queries

Let's now add a query to plot the Pressure Head along a vertical profile. The query will consist of a single vertical line segment, from the vertex at the crest of the slope, to the bottom of the external boundary.

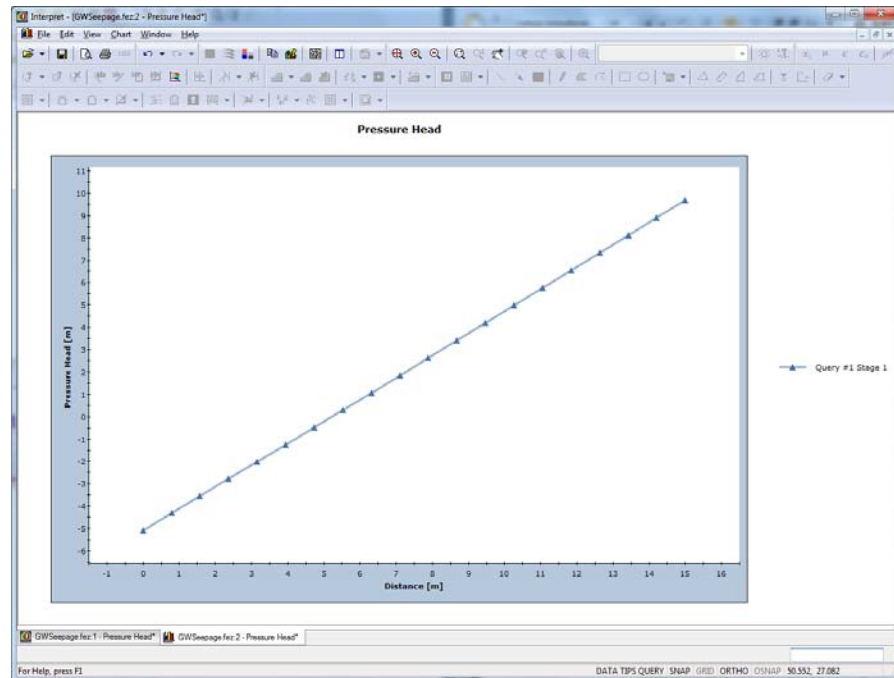


1. Select **Add Material Query** from the toolbar or the Query menu.
2. The Snap option should still be enabled. Click the mouse on the vertex at the crest of the slope, at coordinates (50,35).
3. Enter the coordinates (50,20) in the prompt line, as the second point (or if you have the Ortho Snap option enabled, you can enter this graphically).
4. Right click and select Done, or press Enter. You will see the following dialog.



5. Enter a value of 20 in the edit box. Enable the Show Queried Values checkbox (if it is not already selected). Select OK.
6. The query will be created, as you will see by the vertical line segment, and the display of interpolated values at the 20 points along the line segment.
7. Zoom in to the query, so that you can read the values.
8. We can graph these data with the Graph Material Queries option in the Graph menu or the toolbar. Let's use a shortcut instead.
9. A shortcut to graph data for a single query, is to right click the mouse ON the Query line. Do this now, and select Graph Data from the popup menu.
10. You will see the Graph Query Data dialog. Select the Create Plot button, and the graph will be generated, as shown in the following figure.





Pressure Head profile along vertical section.

The Query we have created gives us the pressure head along a vertical line from the crest of the slope to the bottom of the external boundary. These data are obtained by interpolation from the Pressure Head contours.

Notice the negative Pressure Head (i.e., matric suction) above the Water Table.

Although we have only used a single line segment to define this Query, in general, a Query can be an arbitrary polyline, with any number of segments, added anywhere on or within the external boundary.

Close the graph, and select Zoom All (if you previously zoomed in to read the query values).

Also, delete the Query (right-click on the Query and select Delete Query from the popup menu).

Stress Analysis Results

Let's just have a quick look at the stress analysis results before we conclude this tutorial.



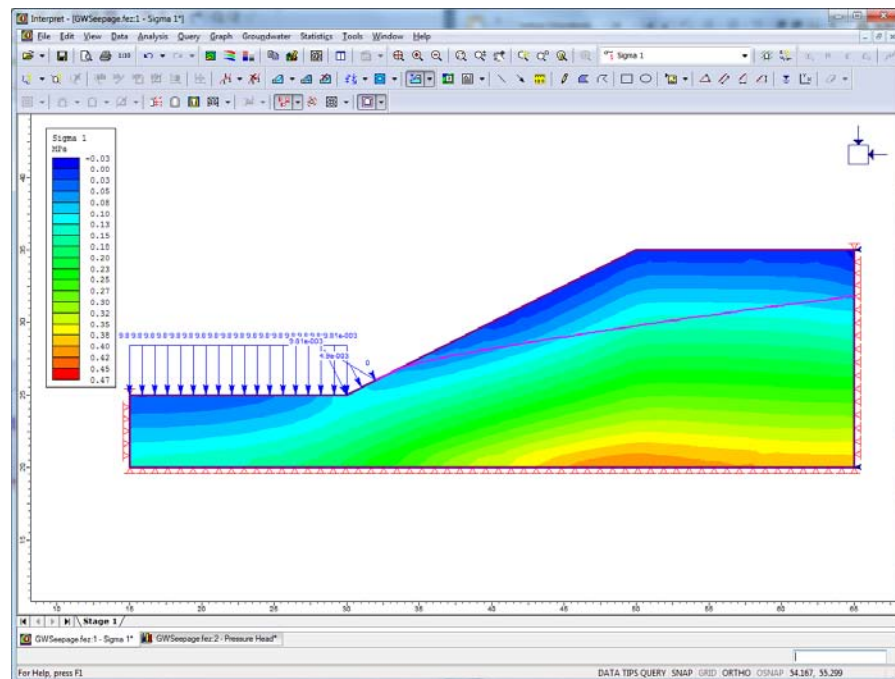
First let's hide the groundwater boundary conditions, and display the stress analysis boundary conditions, by selecting the Show Boundary conditions option (available in the toolbar or the Groundwater menu).

The stress analysis boundary conditions are now displayed. This includes the Fixed X,Y restraints on the external boundary, as well as the distributed load due to Ponded Water (represented by the blue arrows applied normal to the boundary at the toe of the slope).

Because we have computed the stress analysis as well as the groundwater seepage analysis, all of the data from both analyses are available for plotting, by selecting a data type from the drop-list in the toolbar.

For example, you can select the following stress analysis results for plotting:

- Principal stresses (Sigma1, Sigma3, SigmaZ)
- Displacements (Total, Horizontal, Vertical)
- Strength Factor
- Effective Stress



Contours of effective Sigma1

The effective stress results in *Phase2* utilize the pore pressures obtained from the groundwater seepage analysis.

The effective stress results are used in the failure criterion for each material, when computing Strength Factor and yielding.

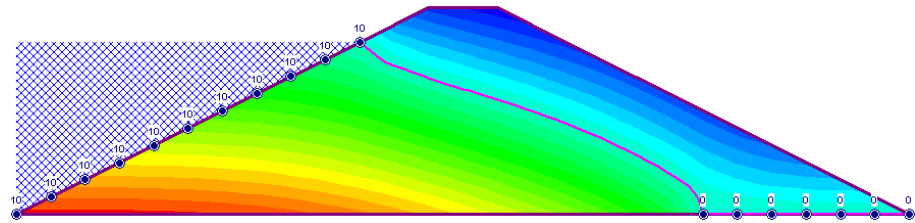
More Groundwater Examples

Many more examples of groundwater seepage analysis with *Phase2* are presented in the *Phase2* Groundwater Verification Manual.

The files used for the verification examples can be found in the Groundwater Verification sub-folder, in the Examples folder in your *Phase2* installation folder.

These examples demonstrate more advanced features of the *Phase2* groundwater analysis, including material permeability functions, infiltration boundary conditions, and other features.

For more information see the Groundwater Verification manual which can be accessed through the *Phase2* Help system.



Pressure head contours in dam with full reservoir